Recover the Design

Degree-Constrained Exact Search with
Event-Driven Propagation and MRV

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Abstract

This project reconstructs an unknown fence design from the degrees of connector positions. Fences are straight horizontal or vertical segments and may pass through empty cells. Two perpendicular fences may cross only when the crossing position contains a connector. The main difficulty is not merely satisfying vertex degrees: every selected long segment also consumes availability at every empty cell it crosses.

We model each possible fence as a binary edge between consecutive non-zero connectors in one row or column. The resulting problem is a degree-constrained selection problem with pairwise conflict constraints. The exact solver combines static infeasibility checks, event-driven forcing, immediate conflict rejection, rollback, and minimum-remaining-values (MRV) branching. A separate validator reconstructs degrees, endpoint agreement, row ordering, and crossings from the final text output.

Correctness was checked by exhaustive small-grid enumeration, randomized differential tests, planted-feasible layouts, and targeted geometric cases. The 24-case Release benchmark uses five measured runs per input. Every case completed within the 30-second safety timeout; the largest median was 332.151 ms. Conflict propagation reduced the largest planted search from 39,888 to 162 nodes, while MRV helped medium inputs but did not improve every case. A separate fixed-seed adversarial planted instance remains pending after a 30-second limit, documenting the search-tree growth that the acceptance corpus does not cover.

Keywords: layout recovery; degree constraints; candidate graph; event-driven propagation; MRV; exact search; algorithm validation

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1 Introduction

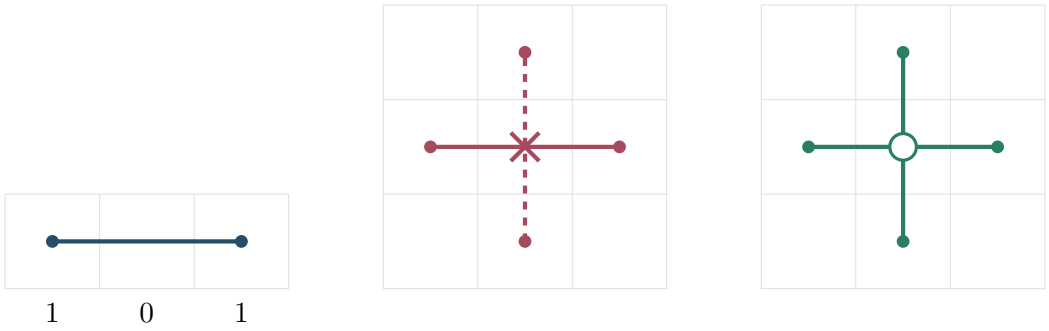
1.1 Problem description

The input is an $n \times m$ rectangular grid of unit-square positions. A positive value $k \in \{1, 2, 3, 4\}$ denotes a connector that must be incident to exactly k fence segments. A zero denotes an empty position. Fences are horizontal or vertical and may be longer than one unit: they may pass through empty positions. Fences cannot bend, and horizontal and vertical fences may cross only at a connector.

For each connector, the program outputs its row, column, and four binary states: up, down, left, and right. If no design exists, the output is `No Solution`. The answer is not unique, so the objective is to find any valid design or to prove that none exists.

1.2 Key modelling observation

An initial simplification might connect only adjacent non-zero cells. Sample 3 shows why this is insufficient: a connector in row 1, column 3 connects to a connector in row 1, column 5, while row 1, column 4 is empty. Thus a candidate fence must be a maximal straight segment between consecutive non-zero connectors in one row or one column.



(a) Long fence across a zero

(b) Illegal empty crossing

(c) Legal connector crossing

Figure 1: Candidate geometry. A long segment may cross empty cells. Perpendicular segments form a conflict when they intersect at an empty position, but crossing at a connector is legal.

1.3 Why the problem is not a plain degree matching

If crossing conflicts were absent, selecting candidate edges to meet prescribed degrees would be a b-matching relaxation. That relaxation is not a complete solution. Candidate intervals in a row are disjoint, but a horizontal interval and a vertical interval can intersect inside a zero cell. Such a pair cannot be selected simultaneously. The conflict relation therefore introduces a global 0-1 exclusion constraint.

The candidate graph is also not guaranteed to be bipartite. A collinear sequence of connectors can create odd cycles after gaps are skipped. Consequently, a simple source-left/right-sink flow construction is not justified by the input geometry.

1.4 Contributions

This implementation contributes:

- a long-segment candidate graph that spans arbitrary empty gaps;
- a symmetric conflict graph built by scanning empty cells;

- an event queue that propagates forced local decisions to fixed point;
- immediate crossing rejection and rollback with state trails;
- MRV branching on the number of local degree-completion combinations;
- an independent validator and a deterministic correctness/performance suite.

2 Data Structure and Algorithm Specification

2.1 Notation and candidate graph

Let V be the set of non-zero connectors and E the set of candidate fence segments. In each row, sort the non-zero columns and add one candidate between each consecutive pair. Perform the symmetric construction in each column. For a connector v, let $E_v \subseteq E$ be its incident candidates. The geometry ensures $|E_v| \leq 4$.

Each candidate stores its two endpoint vertex ids, the direction from each endpoint toward the other, and the original endpoint coordinates. A candidate is horizontal if its endpoint rows are equal.

2.2 Conflict graph

For each pair of a horizontal candidate e and a vertical candidate f, insert a symmetric conflict edge when the two segments intersect strictly inside both open intervals. An equivalent scan visits every empty grid cell and stores at most one horizontal and one vertical candidate there. Since non-overlapping candidate intervals partition each row and column, one empty cell yields at most one conflict pair. The total number of conflict pairs is therefore $O(nm)$.

2.3 Exact constraints

Let $x_e \in \{0, 1\}$ indicate whether candidate e is selected and d_v be the required degree of connector v. Then

$$\sum_{e \in E_v} x_e = d_v \qquad \forall v \in V, \tag{1}$$

$$x_e + x_f \leq 1 \qquad \forall (e, f) \in C, \tag{2}$$

where C is the conflict set. Every selected segment contributes one degree at each endpoint, so exactly $\frac{1}{2}\sum_v d_v$ candidate edges must be selected.

2.4 Static feasibility checks

Before search the model rejects:

- an odd total degree;
- a connector whose required degree exceeds its number of candidate edges;
- a candidate component with odd degree demand or too few candidate edges.

These are necessary conditions. Passing them only means that search is worth starting.

2.5 Event-driven propagation

Each candidate edge has one of three states: Unknown, Selected, or Rejected. For connector v, the solver maintains s_v, the number of selected incident edges, and u_v, the number of unknown

incident edges. An edge-state change updates its two endpoints and appends both connectors to an event queue.

For a queued connector with required degree d_v:

- fail if $s_v > d_v$ or $s_v + u_v < d_v$;
- if $s_v = d_v$, reject every unknown incident edge;
- if $s_v + u_v = d_v$, select every unknown incident edge.

Selecting an edge immediately rejects all of its crossing candidates. Rejections enqueue their endpoints, so propagation continues to a fixed point.

The queue stores connector ids, not edges or assignments. The implementation uses a vector and a moving head to avoid repeated deque relocation. Duplicate events are harmless because each event rereads the current counters.

```
while pending_head < queue_size:
    v = queue[pending_head++]
    if selected[v] > required[v]: return false
    if selected[v] + unknown[v] < required[v]: return false

    if selected[v] == required[v]:
        reject_all_unknown_edges(v)
    else if selected[v] + unknown[v] == required[v]:
        select_all_unknown_edges(v)
```

Listing 1: Forced propagation at one connector

2.6 MRV branching

After propagation, an unsatisfied connector needs

$$need_v = d_v - s_v$$

additional edges. If it has u_v unknown candidates, the number of immediate local assignments is

$$B_v = \binom{u_v}{need_v}.$$

Because $u_v \le 4$, $B_v \le 6$. MRV selects the connector with minimum B_v. Ties prefer larger conflict incidence and then the smaller vertex id.

MRV is a variable-ordering heuristic. It does not restrict the allowed values, so every legal completion remains reachable.

2.7 DFS, rollback, and completeness

The search first propagates, then checks whether every connector has exact degree. Otherwise it enumerates every $need_v$-subset of unknown incident edges for the MRV connector. Selected edges are committed and the other incident unknown edges are rejected.

Every state transition records its edge id and previous state in a trail. Failure rolls back to the branch checkpoint, restores both degree counters, and tries the next subset. Completeness follows because any valid completion must choose one of the enumerated local subsets.

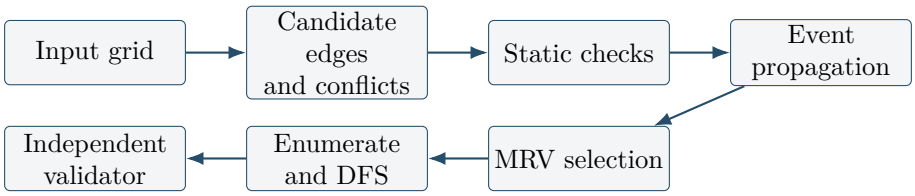


Figure 2: Exact solving pipeline. Propagation reaches a local fixed point before MRV chooses the next connector. A failed branch restores all edge states and counters through the trail.

2.8 Correctness argument

Propagation only derives facts that are necessary in every completion. If a connector has already reached its degree, another selected incident edge would violate it; if all remaining edges are required to reach the degree, none can be rejected. A selected edge excludes every crossing candidate by the problem rule.

At a branching node, every completion must choose exactly $need_v$ edges from the connector's unknown candidates. The DFS enumerates every such subset. Therefore no valid completion is skipped. If all descendants fail, the current node has no solution. Returning on the first success is valid because the problem accepts any solution.

3 Testing Results

3.1 Environment and methodology

Table 1: Experimental environment

Item	Configuration
Platform	Windows, Intel CPU
Compiler	MSVC 19.51, Release for benchmarks
Build	CMake with Visual Studio 18 2026 generator
Correctness	Debug and Release CTest
Benchmark repetitions	one warm-up and five measured runs
Randomness	deterministic generator seeds

Correctness uses two independent existence oracles: the formal solver and a small-instance edge-subset enumerator. Found layouts are serialized through the production IO path and checked by the independent validator. The validator recomputes row-major coverage, degree counts, endpoint agreement, and empty-cell crossings.

Testing is organized as two complementary layers. White-box C++ groups exercise parsing, candidate-model invariants, Validator branches, exhaustive differential enumeration, planted witnesses, propagation counters, and geometric boundaries. Black-box CLI tests invoke the submitted executable through standard input, standard output, exit codes, diagnostic modes, and stderr. Every CTest name identifies one group, so a failure immediately localizes the affected layer instead of reducing the entire suite to a single pass/fail signal.

3.2 Correctness coverage

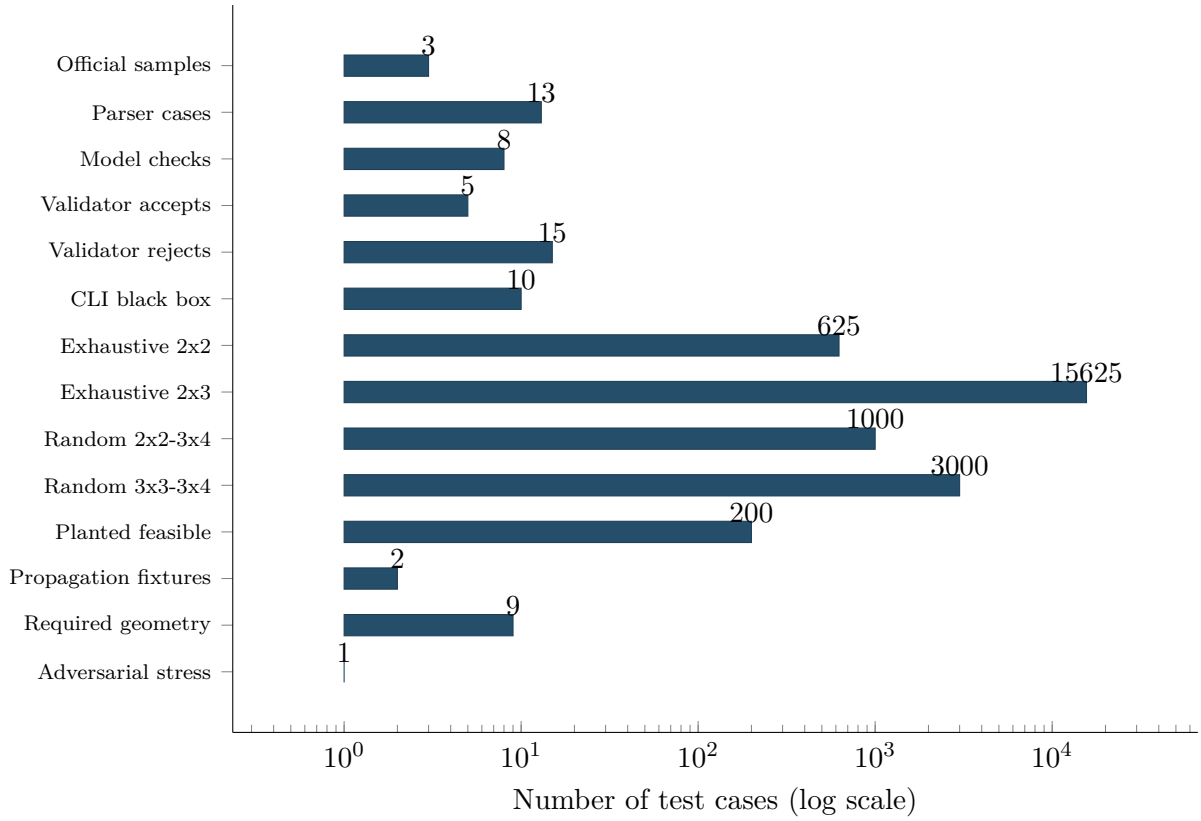


Figure 3: Correctness coverage by test tier. Black-box CLI and rejection groups are listed separately from exhaustive and randomized solver checks.

Table 2: Correctness test cases and observed outcomes

Group	Cases	Purpose	Expected	Observed	Failure diagnosis	Status
Parsing	13	Valid grammar and failure isolation	One success, 12 rejections	All matched	n/a	pass
Model	8	Candidate graph and static proofs	Invariants and reasons hold	All matched	n/a	pass
Validator accepts	5	Valid layouts and empty design	Status `valid`	All matched	n/a	pass
Validator rejects	15	One malformed property per case	Concrete rejection	All matched	n/a	pass
CLI black box	10	Solve, validate, diagnostics, errors	Required streams and codes	All matched	n/a	pass
Official samples	3	Feasible, impossible, non-unique	Accepted result	All matched	n/a	pass
Exhaustive 2×2	625	Every legal degree matrix	Oracle agreement	All matched	n/a	pass
Exhaustive 2×3	15,625	Complete small input space	Oracle agreement	All matched	n/a	pass
Random 2×2–3×4	1,000	Fixed-seed differential	Oracle agreement	All matched	n/a	pass
Random 3×3–3×4	3,000	Larger differential	Oracle agreement	All matched	n/a	pass
Planted feasible	up to 200	Hidden valid witness	Solver finds a layout	All matched	n/a	pass
Propagation	2	Forced and conflict decisions	Counters increase	All matched	n/a	pass
Required geometry	9	Long edges, T, cross, disconnection	Expected feasibility	All matched	n/a	pass

Group	Cases	Purpose	Expected	Observed	Failure diagnosis	Status
Adversarial stress	1	Fixed dense 50 × 50 planted probe	Complete within 30 s	Timeout after 30 s	Search-tree growth	pending

All correctness and acceptance cases pass. The fixed adversarial stress probe is explicitly retained as `pending`: it does not complete within 30 seconds, so it is not reported as a pass or as proof of infeasibility. The exhaustive tests compare existence rather than one output string because multiple valid layouts may exist. For each solver-produced layout the validator checks all constraints independently. Input failures are kept separate from valid but infeasible puzzles, and CLI tests verify that usage and diagnostics never contaminate the answer on standard output.

The default `correctness` CTest label runs the nine named C++ groups, ten CLI cases, and the comment-quality gate when Python is available. The complete 24-case benchmark is registered only when `ADS_PROJ3_ENABLE_PERFORMANCE_TESTS` is enabled and is selected by the separate `performance` label. The adversarial probe similarly lives under the separate `stress` label. This keeps normal correctness runs fast and independent of timing noise.

On the evaluation environment, all 20 correctness-labelled CTest entries passed in both Debug and Release builds. The optional performance entry passed all 24 cases with five deterministic repetitions and no search-metric drift. The stress harness classified its fixed case as `pending` after the 30-second solver limit.

3.3 Benchmark design

The deterministic acceptance corpus contains 24 cases from seed `20261103`. It contains sparse and dense random degree matrices, planted feasible layouts, conflict-heavy grids, fragmented components, and infeasible instances. Sizes range from 10 × 10 to 50 × 50.

Each case is warmed up once. Five measured runs record search nodes, backtracking, propagation events, forced decisions, conflict rejections, depth, and wall time. All structural and search counters are identical across the five repetitions.

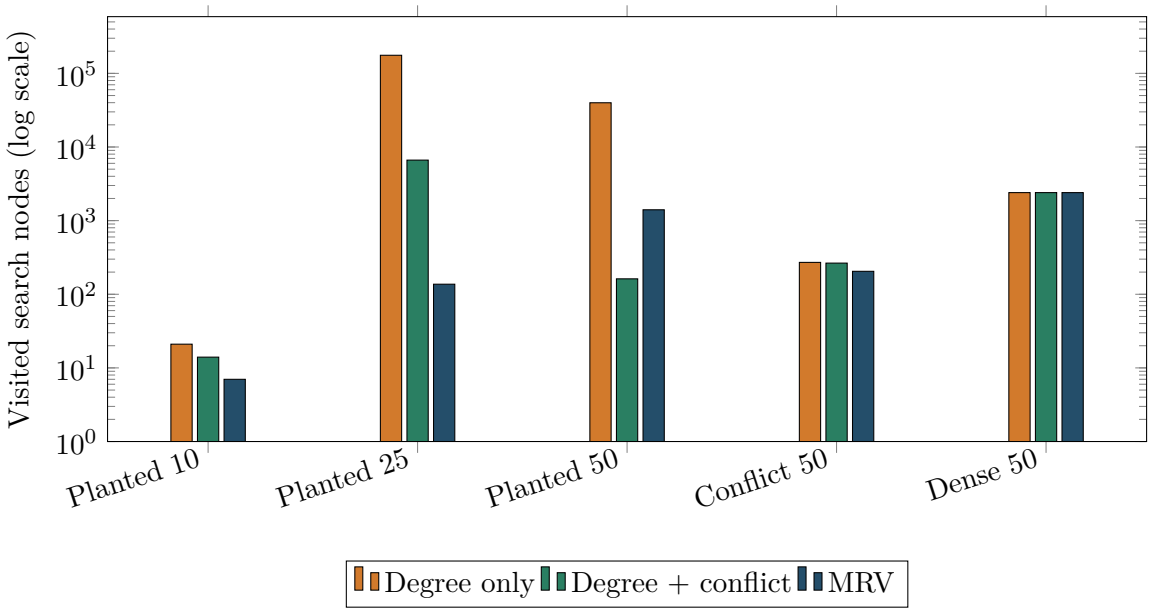


Figure 4: Search-node comparison. Conflict propagation is the dominant reduction; MRV helps the medium planted case but chooses a worse early branch on the largest planted case. The no-propagation baseline is omitted because four large cases timed out.

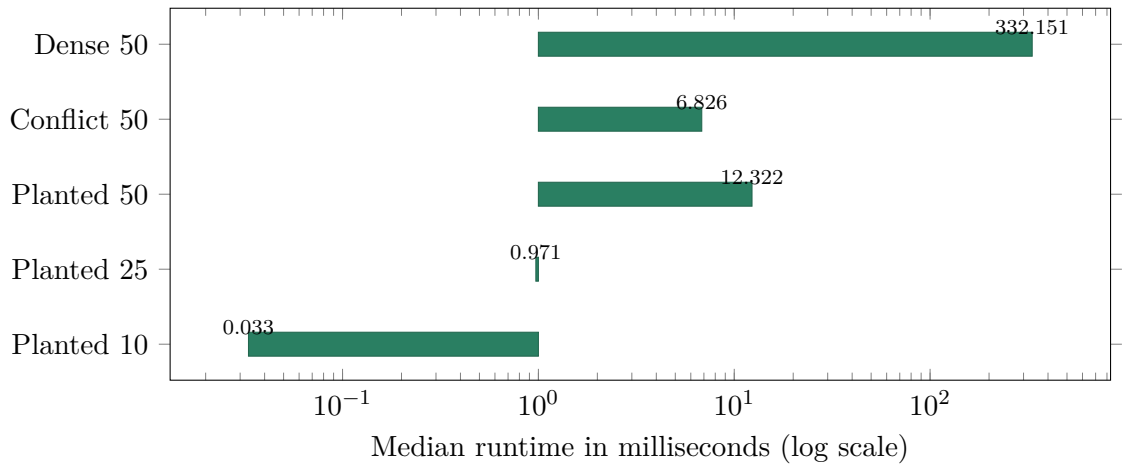


Figure 5: Median Release runtime for representative feasible cases. The dense all-positive graph is slower because MRV scans all unresolved connectors at each node, even though its node count is identical to fixed order.

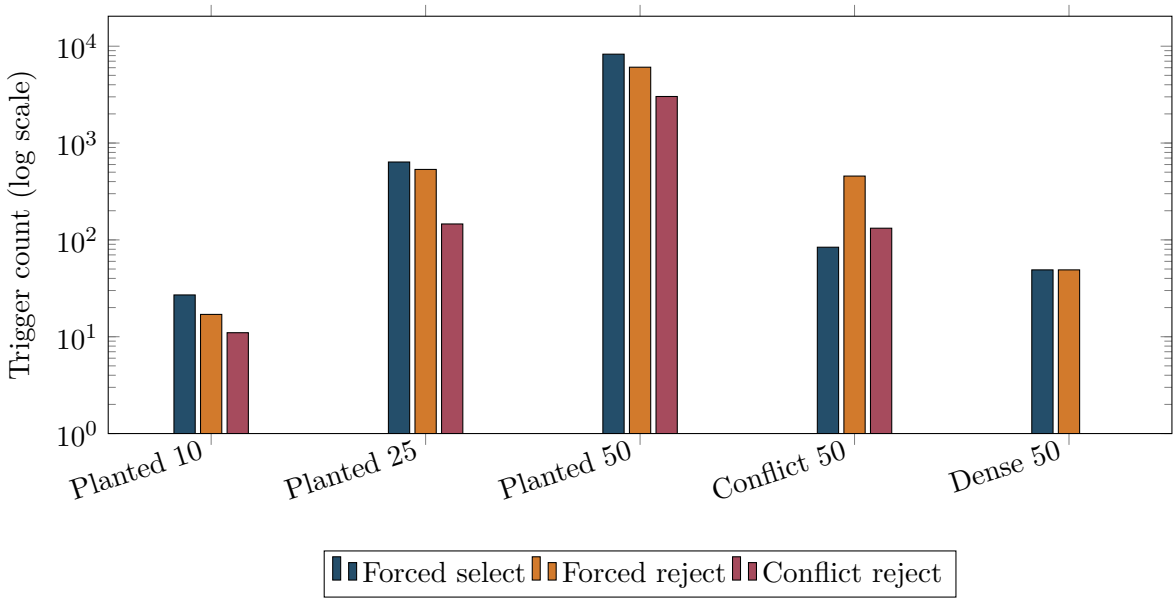


Figure 6: Propagation activity. Forced decisions and crossing rejections account for most of the reduction on planted instances. A zero conflict count is omitted on the logarithmic axis.

3.4 Performance and anomalies

All 24 benchmark cases complete. The maximum median is 332.151 ms and the maximum single run is 349.880 ms. No case approaches the 30-second safety limit.

Table 3: Representative node counts and MRV effect

Case	Degree only	Degree+conflict	MRV	MRV change
Planted 10x10	21	14	7	-50.0%
Planted 25x25	176,118	6,636	137	-97.9%
Planted 50x50	39,888	162	1,404	+766.7%
Conflict-heavy 50x50	271	265	205	-22.6%
Dense all-positive 50x50	2,402	2,402	2,402	0.0%

The no-propagation baseline completes 20/24 cases under a five-second comparison limit. Degree-only propagation completes all 24; conflict propagation further reduces nodes. MRV is not uniformly beneficial. On the largest planted instance, the fixed order happens to choose a very effective early path, while MRV's local minimum-domain choice leads to more total nodes.

This result is useful rather than a failure: it demonstrates that local heuristics must be evaluated against actual search trees. The dense graph also shows that MRV's scan over all connectors can dominate runtime even when it cannot reduce the number of nodes.

4 Analysis and Comments

4.1 Preprocessing complexity

Let P be the number of connectors and M the number of candidate edges. Here $P \le nm \le 2500$ and $M < 5000$. Reading the input and assigning vertex ids costs $O(nm)$. Building row and column candidates is $O(nm + P)$. Marking each candidate through its interior cells costs $O(nm + M)$ in total because each row and column interval is traversed once. The conflict graph has at most one pair per empty cell, hence $O(nm)$ edges.

4.2 Propagation complexity

One connector event examines at most four incident edges. If W edge-state changes occur on one root-to-leaf path, the corresponding event work is $O(W)$ apart from multiple queued events for the same connector. Each such event is constant-sized and repeats only after a relevant counter changes.

4.3 MRV complexity

The current implementation scans all P connectors to choose a branch variable. For each unresolved connector it inspects at most four incident edges and computes a binomial coefficient at most six. One MRV selection is therefore $O(P)$ with a small constant. This explains why dense cases with few backtracks can still spend visible time in repeated MRV scans.

4.4 Search and space complexity

Let N be the number of visited search nodes and L the cost of one node. Total search cost is $O(NL)$; the worst-case number of nodes remains exponential. The algorithm stores one byte or byte-sized state per candidate edge, two counters per connector, a rollback trail, the queue, candidate endpoints, and conflict adjacency. All are bounded by $O(nm + M)$.

4.5 Limitations

No official time limit is given by the problem statement. Measurements therefore describe the current machine, compiler, seed set, and chosen corpus. A separate 50 by 50 planted probe with density 0.50–0.55 exceeded a 15–30 second safety window. It is generated with a fixed seed and retained in the `stress` CTest with status `pending`; it is not part of the all-pass acceptance dataset and is reported as a limitation for dense adversarial degree systems.

The conflict graph captures only forbidden selection pairs. Component decomposition is not applied directly because conflict edges can couple candidate graph components that share no connector.

4.6 Possible improvements

The most direct improvement is to maintain the set of unresolved connectors and incrementally track candidate domain sizes instead of rescanning all connectors for every MRV decision. A second option is one-step lookahead after each local combination. A third is to decompose conflict-connected components and apply separate max-flow or b-matching bounds when a component is bipartite.

These changes should be evaluated against the current baseline. The existing results show that a heuristic can reduce one family of instances while degrading another, so portfolio selection based on model structure may be more useful than applying one ordering rule everywhere.

A Source Code and Reproduction

A.1 Source organization

Table 4: Main source responsibilities

File	Responsibility
`io.cpp/hpp`	Strict input parsing and row-major output
`model.cpp/hpp`	Connector, candidate and conflict construction
`solver.cpp/hpp`	Propagation, rollback, MRV and DFS
`validator.cpp/hpp`	Independent answer validation
`test_runner.cpp`	Nine named unit, solver, and Validator groups
`run_cli_tests.cmake`	Black-box CLI streams and exit codes
`run_performance_tests.cmake`	Optional deterministic corpus CTest
`run_stress_tests.cmake`	Fixed adversarial pending/timeout probe
`benchmark_runner.cpp`	Repetition and CSV instrumentation

A.2 Key source excerpts

The complete source is in the submission `code` directory. The following excerpt shows the two propagation forcing conditions.

```
if (selected == required) {
    // Any additional edge would exceed the target degree.
    for (int edgeId : incidentEdges[v])
        if (state[edgeId] == UNKNOWN)
            assignRejected(edgeId);
} else if (selected + unknown == required) {
    // Every unknown edge is required to reach the target.
    for (int edgeId : incidentEdges[v])
        if (state[edgeId] == UNKNOWN)
            assignSelected(edgeId);
}
```

Listing 2: Forced propagation excerpt

The next excerpt shows the local domain size used by MRV.

```
int need = requiredDegree[v] - selectedDegree[v];
int combinations = 1;
for (int i = 0; i < need; ++i) {
    combinations *= static_cast<int>(available.size()) - i;
```

```
    combinations /= i + 1;
}
// Choose the connector with the smallest value of C(available,need).
```

Listing 3: MRV domain size excerpt

A.3 Build and run

```
cmake -S . -B build -G "Visual Studio 18 2026" -A x64
cmake --build build --config Release
ctest --test-dir build -C Release -L correctness --output-on-failure
./build/Release/recover.exe < input.txt
```

Listing 4: Reproduction commands

B Benchmark Reproduction

The report figures are rendered by PGFPlots from compact CSV files generated by `tools/prepare_report_data.py`. The full results are generated by:

```
python tools/generate_cases.py --output-dir data/generated
python tools/run_benchmarks.py \
  --benchmark-exe build/Release/recover_benchmark.exe \
  --cases-dir data/generated --output-dir data/experiments \
  --label mrv --repeat 5 --timeout 30
python tools/compare_benchmarks.py
python tools/prepare_report_data.py
```

When Python 3 is available, the same 24-case corpus can be run through CTest:

```
cmake -S . -B build -DADS_PROJ3_ENABLE_PERFORMANCE_TESTS=ON
ctest --test-dir build -C Release -L performance --output-on-failure
```

The adversarial stress probe is run separately and records both completion and the expected pending timeout state:

```
cmake -S . -B build -DADS_PROJ3_ENABLE_STRESS_TESTS=ON
ctest --test-dir build -C Release -L stress --output-on-failure
```

Each raw benchmark row contains model size, conflict count, found status, DFS nodes, backtracks, propagation events, forced select/reject counts, crossing rejections, maximum depth, and wall time. The aggregation script reports median, mean, and maximum runtime.

References

[1] S. Russell and P. Norvig. *Artificial Intelligence: A Modern Approach*. Pearson, fourth edition, 2020.

[2] D. E. Knuth. *The Art of Computer Programming, Volume 4A: Combinatorial Algorithms*. Addison-Wesley, 2011.

Author List

Author identities and contribution details are withheld in this anonymous peer review version.

Declaration

We hereby declare that all work done in this project is of our independent effort as a group.